

Interference Mitigation Paradigm for Multibeam Satellites With Hybrid Analog and Digital Arrays: Phase-Only Beam Nulling

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Abstract—With the rapid development of satellite-enabled Internet of Things (IoT), malicious interference has become an inevitable issue, especially in the uplink due to the exposed satellite location. To mitigate uplink interference, this article investigates the beam nulling technique for multibeam satellite communication systems equipped with hybrid analog and digital (HAD) arrays. Unlike fully digital (FD) arrays, the beamforming for HAD architectures consists of a high-dimensional analog part implemented exclusively with phase shifters (PSs) and a low-dimensional digital part with adjustable amplitude and phase, introducing the challenge of limited degrees of freedom. Despite extensive research on HAD beamforming, there is no consensus on the efficient paradigm to implement beam nulling against extremely strong malicious interference. In this article, we first verify that phase-only beam nulling is an achievable paradigm that fully exploits the beam utilization for multibeam satellites while effectively mitigating interference. Additionally, we design two normalized phase-only beam nulling approaches based on majorization minimization and per-antenna optimization, respectively. Simulation results substantiate our analysis concerning the beam nulling paradigm for HAD arrays, and validate the superiority of our proposed approaches which require significantly lower computational complexity. Furthermore, a prototype is developed to demonstrate the practical effectiveness of the proposed phase-only beam nulling approaches.

Index Terms—Hybrid analog and digital (HAD) arrays, interference mitigation, Internet of Things (IoT), multibeam satellite communication systems, phase-only beam nulling.

I. INTRODUCTION

THE Internet of Things (IoT) has become a significant technology to connect billions of devices, enabling reliable data transmission across vast networks [1]. As

the distribution of IoT devices expands to remote regions (e.g., oceans, deserts, and rural areas), terrestrial networks face limitations in coverage and capacity. This challenge has given rise to satellite-enabled IoT, which leverages satellite communication to provide globally ubiquitous connectivity [2], [3], [4], and thereby supports critical IoT applications in environmental monitoring, agriculture, and disaster management [5], [6], [7]. For instance, a multiobjective anticollision mechanism has been developed to address the access-ranging bottleneck in massive IoT connectivity, enabling rapid satellite network formation and thereby mitigating the impact of disasters [8]. On the other hand, driven by the higher-capacity demand, the communication satellites are transitioning from legacy single-beam architectures to multibeam systems which reuse carrier bandwidth across all beams to increase the transmission rates [9], [10], [11], [12]. However, due to the broadcast nature and exposed location, multibeam satellites not only suffer from unavoidable interbeam interference, but also remain vulnerable to external interference caused by malicious jammers [13], [14], [15].

To achieve reliable demodulation at the satellite side, it is essential to adopt effective interference mitigation measures, such as spread spectrum and interference avoidance techniques [16], [17], [18], [19]. While these techniques effectively suppress interference, they do so at the expense of spectral efficiency, inevitably resulting in decreased data rates. Beyond these conventional measures, beam nulling offers a promising alternative by exploiting the spatial domain to form deep notches steered toward jammer directions, thereby significantly reducing interference reception gain without sacrificing spectral efficiency [20], [21]. Consequently, beam nulling provides a superior solution for interference mitigation in multibeam satellite communication systems. It is worth noting that phased array antennas have emerged as the preferable architecture for forming multiple spot beams [22], which naturally facilitates the implementation of beam nulling techniques.

Currently, most satellite communication systems have migrated to Ka or Q/V band to achieve wider spectrum bandwidth and higher transmission rates [23], therefore requiring large-scale antenna arrays (LSAAs) to combat the severe propagation loss and atmospheric attenuation. It should be noted that the implementation of LSAAs with fully digital (FD) structures is systematically complicated and prohibitively

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expensive when the number of antenna elements reaches hundreds or even thousands, since FD arrays require each antenna element to be connected to an independent radio frequency (RF) chain [24]. In contrast, hybrid analog and digital (HAD) architectures, which connect large numbers of antennas to significantly fewer RF chains through analog beamforming networks, are more economically and operationally practical, offering a preferable architectural solution for LSAAAs [25].

Hybrid beamforming (HBF) for HAD arrays has received extensive research over the past decade, originally applied in 5G massive MIMO systems [26]. Previous research on HBF can be categorized into two distinct groups tailored to single-user (SU) and multiuser (MU) scenarios. In terms of SU-HBF, the most typical approach is matrix factorization which minimizes the Frobenius norm between the HBF matrix and the fully digital beamforming (DBF) matrix [27], [28]. Note that the analog part of HBF is implemented by phase shifters (PSs) exclusively, therefore introducing constant-modulus constraints in the matrix factorization problem and rendering it remarkably challenging. Another SU-HBF manner employs the minimum mean square error (MMSE) criterion to achieve desired signal detection [29], [30]. It is worth noting that the MMSE problem is equivalent to the spectral efficiency maximization problem [31], and it can be further proven equivalent to a weighted matrix factorization problem [27]. In the MU-HBF scenario, the primary challenge lies in mitigating the unavoidable interuser interference. Most existing studies formulate MU-HBF optimization as a sum-rate maximization problem, which is typically solved using alternating optimization or coordinate descent methods [32], [33], [34], [35].

Beam nulling is a special application of beamforming techniques, and it has been extensively investigated in FD arrays [36]. Actually, from the perspective of HAD arrays, both SU-HBF and MU-HBF can be conceptualized as generalized forms of beam nulling that reduce reception gain in the directions of undesired data streams or interfering users. Whereas, these techniques exhibit limited interference mitigation performance when confronted with strong malicious jammers. In contrast, phase-only beam nulling, which is implemented entirely through phase-shifters at the analog front end, offers a fundamentally different approach by strategically placing precise beam notches directly toward jammer directions. This enables significant suppression of interference signals at the RF stage before they enter the digital processing chain, thereby providing substantially enhanced protection against high-power malicious interference [37], [38], [39], [40], [41], [42], [43], [44]. Adaptive approaches for phase-only beam nulling have garnered extensive investigation [37], [38], [39], attributed to their inherent capability of suppressing interference without explicitly requiring prior knowledge of jammer directions. In this context, adaptive beamforming leverages the statistics of received signals to automatically adjust the complex weights applied to each antenna element, thereby maximizing the signal-to-interference-plus-noise ratio (SINR) of the combined array output. However, these methods necessitate a closed-loop architecture in which each antenna element must be equipped with an independent RF chain, rendering them impractical for HAD arrays due to pro-

hibitive hardware constraints. Conversely, phase-only beam nulling methods based on pattern synthesis remain viable and practical for HAD arrays. These synthesis-based approaches directly form beam patterns with precisely positioned notches by incorporating directional information acquired from well-established direction-of-arrival estimation techniques [45], [46], [47], thereby providing a structurally compatible interference mitigation paradigm for resource-constrained HAD array implementations. Semidefinite relaxation is a commonly used technique for phase-only beam nulling [40], [41], but it is unsuitable for LSAAAs since the computational complexity grows exponentially with the array scale. Gradient projection (GP) provides an alternative feasible method [42], [43], albeit with slow convergence for LSAAAs. Polygons construction is a recently proposed phase-only beam nulling approach, but it is restricted to single-source interference mitigation [44]. Additionally, various hardware impairments in practical systems, such as mutual coupling, amplitude-phase mismatches and component nonlinearity, can cause uncertain beam distortions and exert a nonnegligible impact on interference suppression performance [48], [49], [50], [51]. Although mitigating these impairments in large-scale hybrid arrays remains an important and challenging topic, we focus on the fundamental algorithmic design and performance analysis of phase-only beam nulling under idealized hardware assumptions in this work, which serves as a necessary first step toward more practical implementations.

In this article, we first analyze the performance of existing HBF methods for beam nulling application, and reveal their limitation in beam utilization as well as their inability to combat extremely strong malicious interference. Subsequently, we highlight that phase-only beam nulling is the preferable interference mitigation paradigm for multibeam satellite communication systems equipped with HAD arrays. Furthermore, we propose two normalized phase-only beam nulling algorithms, in which an automatic normalization factor is introduced to enhance the algorithms' general applicability. The first algorithm, based on the majorization-minimization framework, incorporates a bisection mechanism to accelerate convergence, while the second one derives a closed-form solution for per-antenna optimization, both achieving significantly reduced computational complexity. The main contributions and innovations of this work are listed as follows.

- 1) We summarize the existing hybrid analog-digital array beamforming paradigms, and analyze their performance when applied to beam nulling. Our analysis manifests both the feasibility and limitation of these methods for beam nulling application.
- 2) We reveal that phase-only beam nulling emerges as the preferable spatial interference mitigation paradigm for multibeam systems equipped with HAD arrays, offering superior beam utilization efficiency and enhanced resilience to RF chain saturation.
- 3) We propose two phase-only beam nulling algorithms for large-scale hybrid arrays based on majorization minimization and per-antenna optimization. Both algorithms are formulated in a normalized manner, which not only

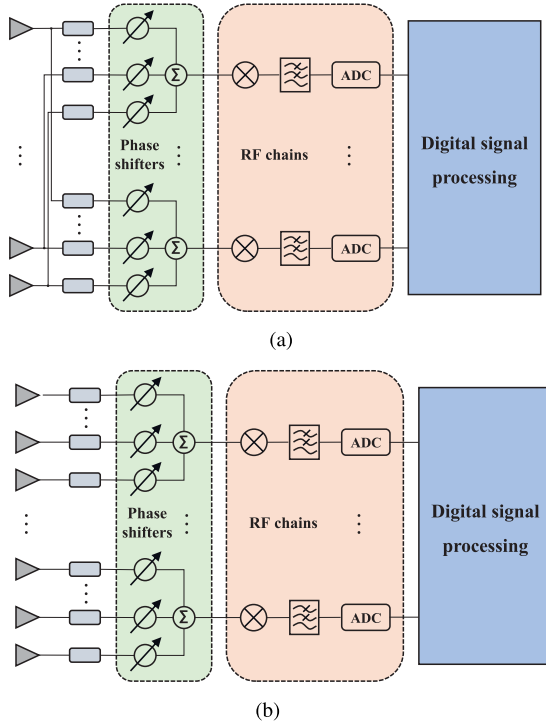


Fig. 1. HAD array structures. (a) Fully connected structure. (b) Partially connected structure.

enhances generalizability across diverse array configurations but also reduces computational complexity.

- 4) We develop a proof-of-concept phased array prototype and implement the proposed phase-only beam nulling algorithms on it to evaluate their performance. The experimental results demonstrate the effectiveness of the proposed algorithms under realistic hardware constraints.

The rest of this article is organized as follows. Section II presents the system model. We analyze the interference mitigation performance of typical HBF paradigms for multibeam systems in Section III, and propose two normalized phase-only beam nulling algorithms in Section IV. Section V details the simulation and experimental results, followed by the conclusions in Section VI.

Notations: Throughout this article, we use lower-case and upper-case boldface characters to represent vectors and matrices, respectively. The superscripts $(\cdot)^T$, $(\cdot)^*$, $(\cdot)^H$ and $(\cdot)^\dagger$ denote the transpose, complex conjugation, conjugate transpose and pseudo inverse, respectively. The notation $\text{tr}(\cdot)$ represents the trace operator of the inner matrix and $\arg(\cdot)$ represents the phase of the inner vector. We use $[\cdot]_m$ to represent the m th element of the inner vector, and $[\cdot]_{m,n}$ to represent the m th row and the n th column of the inner matrix.

II. SYSTEM MODEL

We consider a satellite communication system equipped with an $N_x \times N_y$ rectangular HAD array, where N_x elements are arranged along the x -axis and N_y elements are positioned along the y -axis. In terms of the HAD structures illustrated in Fig. 1, $N_x N_y$ antenna elements are connected to M RF chains ($M \ll N_x N_y$), where the fully connected structure represents

a co-array architecture and the partially connected structure corresponds to a subarray form. In this regard, suppose L to be the number of practically operational beams for a multibeam satellite communication system. It is important to note that the maximum number of simultaneously achievable beams is fundamentally constrained by the number of available RF chains in the HAD architecture, i.e., $L \leq M$. This hardware limitation arises from the fact that each RF chain can only support one independent data stream in spatial domain, thereby establishing a direct relationship between the number of RF chains and multibeam capacity. When $L < M$, the multibeam capability is underutilized. The analysis presented in this article is applicable to both fully connected and partially connected HAD structures. For clarity and conciseness in the subsequent presentation, we focus on the fully connected HAD structure without loss of generality.

For a satellite-borne rectangular array, the received signal from a ground-based station before beamforming is given by

$$\mathbf{x}(t) = \mathbf{g}s(t) + \mathbf{n}_w(t) \quad (1)$$

where $s(t)$ is the transmitted signal, $\mathbf{n}_w(t)$ is the additive white Gaussian noise, and $\mathbf{g} \in \mathbb{C}^{N_x N_y}$ is the corresponding channel vector. According to [52], [53], and [54], such a channel vector can be typically represented as

$$\mathbf{g} = \sqrt{L_f} e^{j\varphi} \mathbf{a}(\theta, \phi) \quad (2)$$

where φ is the propagation phase with uniform distribution over $[0, 2\pi)$, L_f is the propagation loss including large-scale fading and small-scale attenuation, and $\mathbf{a}(\theta, \phi)$ is the array steering vector. According to [55], the steering vector of a rectangular array can be expressed as

$$\mathbf{a}(\theta, \phi) = \mathbf{a}_x(\theta, \phi) \otimes \mathbf{a}_y(\theta, \phi) \quad (3)$$

where $\otimes$ represents the Kronecker product, θ and ϕ are the elevation angle and the azimuth angle, and

$$\begin{aligned} \mathbf{a}_x(\theta, \phi) &= \left[1, e^{j\frac{2\pi}{\lambda} d_x \sin \theta \cos \phi}, \dots, e^{j\frac{2\pi}{\lambda} (N_x-1) d_x \sin \theta \cos \phi} \right]^T \\ \mathbf{a}_y(\theta, \phi) &= \left[1, e^{j\frac{2\pi}{\lambda} d_y \sin \theta \sin \phi}, \dots, e^{j\frac{2\pi}{\lambda} (N_y-1) d_y \sin \theta \sin \phi} \right]^T \end{aligned} \quad (4)$$

where d_x and d_y represent the element spacing along the x - and y -axis, respectively. Suppose that K interference sources are located at $\{(\theta_l^k, \phi_l^k)\}_{k=1}^K$, and the l th beam of the multibeam satellite points to (θ_l^l, ϕ_l^l) , $l = 1, \dots, L$. We construct the manifold matrix of the l th beam as $\mathbf{A}_l = [\mathbf{a}(\theta_l^l, \phi_l^l), \mathbf{a}(\theta_l^1, \phi_l^1), \dots, \mathbf{a}(\theta_l^K, \phi_l^K)]$, which provides a structured representation of the satellite channel by arranging the steering vectors associated with the different incident directions of the corresponding transmitters.

In terms of the HAD architecture, we denote $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_M] \in \mathbb{C}^{N_x N_y \times M}$ as the high-dimensional analog part of the HBF matrix, and $\mathbf{F} = [\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_L] \in \mathbb{C}^{M \times L}$ as the low-dimensional digital part. Note that the analog part is implemented using PSs, thereby imposing a constant-modulus constraint on each element of $\mathbf{W}$. The overall beamforming vector for the l th beam can be equivalently formulated as $\mathbf{W} \mathbf{f}_l$, followed by the array response of desired directions as $\mathbf{A}_l^H \mathbf{W} \mathbf{f}_l$, $l = 1, \dots, L$.

III. BEAM NULLING PARADIGM ANALYSIS

Existing research on HBF consistently strives to approximate the performance of DBF. In this section, we analyze the performance of typical HBF paradigms when applied to beam nulling, and reveal their feasibility as well as deficiency for multibeam satellite communication systems, especially in the case of strong malicious interference.

A. Case for $L \leq \frac{M}{2}$

The conventional approach for HBF first designs an unconstrained DBF matrix, and then decomposes it into a constant-modulus part for high-dimensional analog beamforming as well as an unconstrained part for low-dimensional digital beamforming. To implement beam nulling, the unconstrained DBF vector $\mathbf{v}_l$ for the l th beam can be calculated through

$$P_1 : \min_{\mathbf{v}_l \in \mathbb{C}^{N_x N_y}} \rho_l \quad (5a)$$

$$\text{s.t. } |\mathbf{a}^H(\theta_B^l, \phi_B^l) \mathbf{v}_l| \geq \rho_B \quad (5b)$$

$$|\mathbf{a}^H(\theta_I^k, \phi_I^k) \mathbf{v}_l| \leq \rho_I, k = 1, \dots, K \quad (5c)$$

where (5b) ensures that the main-lobe gain of the beam pointing to (θ_B^l, ϕ_B^l) is higher than ρ_B , and (5c) keeps the array response in the undesired directions (θ_I^k, ϕ_I^k) , $k = 1, \dots, K$ lower than ρ_I . The DBF problem P_1 aims to minimize the reception gain in the directions of interference sources, therefore mitigating the impact of interference. By stacking the beamforming vector for each beam, we can represent the overall DBF matrix as $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_L] \in \mathbb{C}^{N_x N_y \times L}$.

In the case where the number of RF chains is greater than or equal to twice the number of operational beams, it has been demonstrated that the HAD arrays can achieve the same beamforming performance as FD arrays, i.e., there exist $\mathbf{W}$ and $\mathbf{F}$ such that $\mathbf{V} = \mathbf{W}\mathbf{F}$ [33].

Proposition 1: When $L = M/2$, the closed-form HBF matrices $\mathbf{W}$ and $\mathbf{F}$ which satisfy $\mathbf{V} = \mathbf{W}\mathbf{F}$ can be formulated as

$$[\mathbf{F}]_{2l-1,l} = [\mathbf{F}]_{2l,l} = \frac{1}{2} \max_n |[\mathbf{V}]_{n,l}| \quad (6a)$$

$$[\mathbf{F}]_{m,l} = 0, m \neq 2l-1 \ \& \ m \neq 2l \quad (6b)$$

$$[\mathbf{W}]_{n,2l-1} = e^{j(\arg([\mathbf{V}]_{n,l}) - \cos^{-1}([\mathbf{V}]_{n,l}/2[\mathbf{F}]_{2l-1,l}))} \quad (6c)$$

$$[\mathbf{W}]_{n,2l} = e^{j(\arg([\mathbf{V}]_{n,l}) + \cos^{-1}([\mathbf{V}]_{n,l}/2[\mathbf{F}]_{2l,l}))} \quad (6d)$$

where $n = 1, \dots, N_x N_y$, $m = 1, \dots, M$ and $l = 1, \dots, L$.

Herein, we omit the proof details of Proposition 1 which can be found in [33]. Furthermore, when $L < M/2$, we can represent the two HBF matrices as $\mathbf{W} = [\mathbf{W}_s, \mathbf{W}_r]$ and $\mathbf{F} = [\mathbf{F}_s^T, \mathbf{0}_{(M-2L) \times L}^T]^T$ where $\mathbf{W}_s \in \mathbb{C}^{N_x N_y \times 2L}$ and $\mathbf{F}_s \in \mathbb{C}^{2L \times L}$ can be calculated through Proposition 1 since the condition of their dimensions is satisfied. Meanwhile, $\mathbf{W}_r \in \mathbb{C}^{N_x N_y \times (M-2L)}$ can be any arbitrary matrix that satisfies the unit-modulus constraint, as the residual elements of $\mathbf{F}$ that correspond to $\mathbf{W}_r$ are all zero.

Although the HAD architecture with $L \leq M/2$ can achieve the optimal beam nulling performance as the FD architecture,

the effective number of beams is less than or equal to half of the maximum number of achievable beams, resulting in at least half of the capacity loss. This fundamental limitation highlights an inherent tradeoff in this HBF approach: while perfect interference suppression is theoretically possible, it necessitates a substantial sacrifice in beam efficiency. Consequently, the HAD-based multibeam system requires careful balance between interference mitigation performance and beam utilization efficiency.

B. Case for $\frac{M}{2} < L \leq M$

To exploit the potential of beam utilization, the HAD architecture with $M/2 < L \leq M$ is substantially investigated, especially in the case where $L = M$. When it comes to the SU multistream scenario, the HBF problem in existing research is typically modeled as

$$P_2 : \min_{\mathbf{W}, \mathbf{F}} \|\mathbf{\Gamma}(\mathbf{V} - \mathbf{W}\mathbf{F})\|_F \quad (7a)$$

$$\text{s.t. } |[\mathbf{W}]_{n,m}| = 1 \quad \forall n, m \quad (7b)$$

where $\|\cdot\|_F$ represents the Frobenius norm, $\mathbf{\Gamma}$ is a weighted matrix which is set to the identity matrix $\mathbf{I}$ in the spectral efficiency maximization manner [27], [28], while is strategically designed in the MMSE manner [29], [30]. Note that the expression of HBF problem in [29] and [30] and (7) may not be the same, but the equivalence between them has been proved in [31]. To achieve beam nulling for each independent beam, the optimal DBF matrix $\mathbf{V}$ can still be calculated through the optimal beam nulling problem P_1 , as described in (5). Herein, the optimal solutions of P_2 are denoted as $\mathbf{W}^*$ and $\mathbf{F}^* = [\mathbf{f}_1^*, \mathbf{f}_2^*, \dots, \mathbf{f}_L^*]$. We further introduce the following proposition to demonstrate that the HBF paradigm through analog-digital matrix factorization, as described in P_2 , is not suitable for beam nulling against strong malicious interference due to the increased reception gain in the interference direction.

Proposition 2: For the L available beams generated by P_1 and P_2 , there must exist $l' \in \{1, 2, \dots, L\}$ that satisfies the following inequality

$$|\mathbf{a}^H(\theta_I^k, \phi_I^k) \mathbf{W}^* \mathbf{f}_{l'}^*|^2 \geq |\mathbf{a}^H(\theta_I^k, \phi_I^k) \mathbf{v}_{l'}|^2. \quad (8)$$

Proof: Please see Appendix A.

To intuitively demonstrate the phenomenon described in Proposition 2, we plot the beam pattern of one beam in Fig. 2(a), from which the normalized reception gain after matrix factorization rises by 26.17 dB, despite the matrix fitting error being small. Proposition 2 implies that the increased reception gain of such SU-HBF paradigm will cause a reduced SINR performance, further resulting in lower spectral efficiency or potentially causing communication outage when the interference power is sufficiently strong. Consequently, in the case of $M/2 < L \leq M$, we do not recommend this matrix factorization paradigm to design the beam nulling matrix for HAD-based multibeam satellite communication systems.

On the other hand, when it comes to the MU single-stream manner, the HBF problem for an uplink receiver is typically

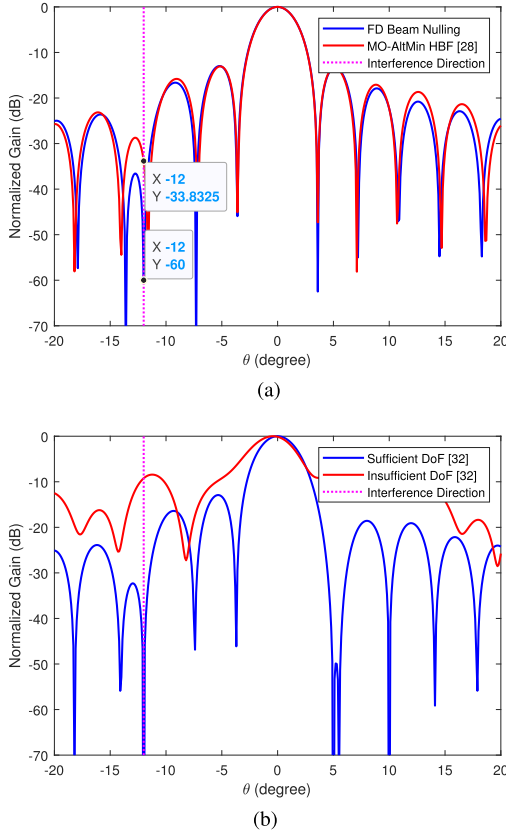


Fig. 2. 1-D pattern slice of one beam in a multibeam system, with a jammer located at 12° . (a) SU-HBF paradigm. (b) MU-HBF paradigm.

formulated as the following optimization problem to maximize the sum rate of all users [32], [33], [34]

$$P_3 : \max_{\mathbf{W}, \mathbf{F}} \sum_{l=1}^L \alpha_l \log_2(1 + \gamma_l) \quad (9a)$$

$$\text{s.t. } |[\mathbf{W}]_{n,m}| = 1 \quad \forall n, m \quad (9b)$$

where α_l represents the priority of the l th beam, and the SINR of the l th beam γ_l is given by

$$\gamma_l = \frac{P_U^l |\mathbf{a}^H(\theta_B^l, \phi_B^l) \mathbf{W} \mathbf{f}_l|^2}{\sigma_n^2 + \sum_{l' \neq l} P_U^{l'} |\mathbf{a}^H(\theta_B^{l'}, \phi_B^{l'}) \mathbf{W} \mathbf{f}_l|^2 + \sum_k P_I^k |\mathbf{a}^H(\theta_I^k, \phi_I^k) \mathbf{W} \mathbf{f}_l|^2} \quad (10)$$

Herein, σ_n^2 is the noise power, the second term of denominator is the interbeam interference where P_U^l denotes the arriving power of the l th user, and the third term is the external malicious interference where P_I^k denotes the arriving power of the k th jammer. Actually, the third term has not been considered in existing research, but we include it for beam nulling analysis. From the second and third terms in the denominator of (10), it can be discovered that the malicious interference can be equivalently considered as interbeam interference with the power of P_I^k and the direction at (θ_I^k, ϕ_I^k) , $k = 1, \dots, K$. Under this MU-HBF paradigm, external interference will be treated as interbeam interference and then suppressed. To solve the challenging MU-HBF problem P_3 , a commonly used approach is to alternately obtain $\mathbf{W}$ and $\mathbf{F}$ through power optimization (PO) and zero forcing (ZF) respectively. In this regard, PO for $\mathbf{W}$ can enhance the reception gain for each independent beam, and ZF for $\mathbf{F}$ can facilitate forming deep notches toward

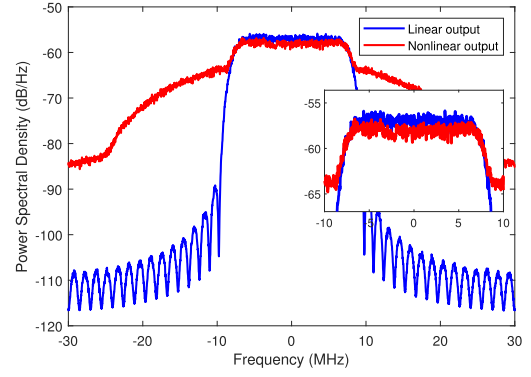


Fig. 3. Power spectral density of the desired signal within linear and nonlinear outputs.

other undesired directions to mitigate interbeam or external malicious interference.

However, the performance of ZF is constrained by the degree-of-freedom (DoF) which represents the maximum number of arbitrarily processable signals [55]. The DoF equals the number of antennas in FD arrays, whereas it equals the number of RF chains in HAD arrays. Therefore, the ZF to achieve beam nulling is effective only when $L + K \leq M$. In the case where $M/2 < L \leq M$, the number of suppressible external interference is $K' = M - L$ when $L < M$, and this HBF paradigm fails to form notches toward external malicious interference when $L = M$, resulting in significant beam distortions. If we consider the feasible case where $L + K \leq M$, with a given $\mathbf{W}$, the ZF expression of $\mathbf{F}$ can be expressed as

$$\mathbf{F} = \mathbf{W}^H \mathbf{\Phi} (\mathbf{\Phi}^H \mathbf{W} \mathbf{W}^H \mathbf{\Phi})^{-1} \mathbf{\Xi}^{\frac{1}{2}} \quad (11)$$

where $\mathbf{\Phi} = [\mathbf{a}(\theta_B^l, \phi_B^l), \dots, \mathbf{a}(\theta_B^L, \phi_B^L), \mathbf{a}(\theta_I^1, \phi_I^1), \dots, \mathbf{a}(\theta_I^{K'}, \phi_I^{K'})]$, and $\mathbf{\Xi}$ is a diagonal matrix with the diagonals as $[P_U^1, \dots, P_U^L, P_I^1, \dots, P_I^{K'}]$. Furthermore, the alternate PO manner to calculate $\mathbf{W}$ is described in [32], [33], and [34], while a low-complexity solution given by [32] straightforwardly represents the analog beamforming matrix $\mathbf{W}$ as $\mathbf{\Phi}^*$.

Under this framework, the beam pattern comparison between the sufficient DoF and the insufficient DoF is depicted in Fig. 2(b), in which beam distortion occurs in the insufficient case where $L + K > M$ ($K = 1$, $L = M = 4$). In summary, the MU-HBF can achieve superior interference mitigation performance by reducing the number of activating beams. Therefore, this HBF paradigm fails to fully exploit the potential beam efficiency while implementing beam nulling, especially in the scenario with excessive interference sources.

C. Case for RF Chain Saturation

On the receiving side, RF components, including low-noise amplifiers (LNAs), PSs, mixers, intermediate frequency filters (IFFs) and analog-to-digital converters (ADCs), predominantly exhibit nonlinear characteristics. Once input power reaches the nonlinear region of RF components, the output signal experiences severe distortion which can be mathematically expressed as [56], [57], [58]

$$x_o(t) = \alpha_1 x_i(t) + \alpha_2 x_i(t)|x_i(t)| + \dots + \alpha_q x_i(t)|x_i(t)|^{q-1} \quad (12)$$

where $x_i(t)$ and $x_o(t)$ are input and output signals respectively, and $\alpha_1, \alpha_2, \dots, \alpha_q$ are coefficients that depend on the input power. It is worth noting that even-order terms can be explicitly filtered out, while odd-order terms typically remain within the desired band. According to the Busgang theorem [59], the nonlinear output (12) can be further decomposed as

$$x_o(t) = g_\alpha x_i(t) + n_d(t) \quad (13)$$

where g_α is the gain when working in the nonlinear region, and $n_d(t)$ is the nonlinear component which is uncorrelated to both $x_i(t)$ and $x_o(t)$. On the one hand, g_α is lower than the ideal linear gain [57]. On the other hand, $n_d(t)$ is equivalent to additional interference, thereby degrading the SINR performance. This distortion phenomenon is illustrated in Fig. 3, where the power spectral density of the desired signal is severely deteriorated in the nonlinear case.

RF saturation occurs when extremely strong malicious interference causes any component to work beyond the linear region [60], [61], and the high-order distortion terms will significantly degrade the demodulation performance. In this case, where $P_I^k \gg P_U^l$, we exclusively focus on the power of malicious interference to analyze the link budget. The arriving power in decibel can be expressed as $P_I^k = P_T^k - L_f^k$ where P_T^k is the transmitting power and L_f^k is the propagation loss including free space loss and rain attenuation. Referring to the system structure illustrated in Fig. 1, we denote the input power of LNA, PS, mixers, IFF and ADC as $P_{LNA}^k, P_{PS}^k, P_{MIX}^k, P_{IFF}^k$ and P_{ADC}^k respectively. Moreover, with the gain of each component, we have $P_{LNA}^k = P_I^k + G_{ANT}$, $P_{PS}^k = P_{LNA}^k + G_{LNA}$, $P_{MIX}^k = P_{PS}^k + G_{ABF}(\theta_I^k, \phi_I^k)$, $P_{IFF}^k = P_{MIX}^k + G_{MIX}$, and $P_{ADC}^k = P_{IFF}^k + G_{IFF}$ in decibel. Note that G_{ANT} , G_{PS} , G_{MIX} and G_{IFF} are all constant because these components apply uniform treatment to input signals, while $G_{ABF}(\theta_I^k, \phi_I^k)$ is spatially selective depending on the analog beamforming matrix $\mathbf{W}$. According to the component parameters released by Analog Devices, Inc., the saturation power levels of these RF components practically belong to the range of [0, 40] dBm [62]. We continue our analysis by studying a specific case, in which the satellite is at the altitude of 1000 kilometers and operating at the Ka-band. Furthermore, suppose that there is a large-scale jammer equipped with a 3-meter antenna and 1000-watt power amplifiers (PAs). Even though the effective isotropic radiated power (EIRP)¹ can reach 87.26 dBW in total, the arriving interference power P_I^k is only -64.72 dBm. The enormous gap from the saturation threshold demonstrates that the RF front end, including LNA and PS, will hardly be saturated by interference. However, the interference power can be sequentially amplified by LNA, analog beamforming, and even the subsequent components within the RF chains. Given an analog beamforming gain of 45 dB and an LNA gain of 25 dB, the resulting interference power at the mixer input reaches 5.28 dBm. Further assuming an overall RF-chain gain of 20 dB, the interference power before digitization becomes 25.28 dBm. According to representative datasheets of satellite-grade RF components [64], [65], [66], the saturation

thresholds of mixers and ADCs are typically on the order of 10 dBm, indicating a high risk of RF saturation under strong interference.

To avoid RF chain saturation, from the perspective of beam nulling, it is necessary to reduce the interference power to a nonthreatening level before combining to the RF chains, which only depends on the analog beamforming network as shown in Fig. 1. Nevertheless, the existing HBF paradigms include analog and digital stages, which finally implements beam nulling in the stage of digital processing with the analog-digital matrix $\mathbf{WF}$. It is implied that if saturation occurs at the analog stage, the signals entering the digital stage will be distorted as well, rendering beam nulling with $\mathbf{WF}$ futile. Therefore, to deal with the extremely powerful malicious interference, we do not recommend the existing HBF paradigms. In contrast, we should significantly reduce the input power to the RF chains, and thus resort to phase-only beam nulling, which employs exclusively the analog beamforming matrix $\mathbf{W}$ implemented by PSs to suppress malicious interference at the RF front end.

IV. PROPOSED NORMALIZED BEAM NULLING APPROACH

In this section, we propose two phase-only beam nulling approaches for multibeam satellite communication systems equipped with HAD arrays. The proposed approaches strategically introduce an automatic normalization factor to remove the dependency on array-specific parameter selection that affects beamforming performance, thereby enhancing their generalizability across diverse array configurations. The first algorithm, based on the majorization-minimization framework, incorporates a bisection mechanism to accelerate convergence, while the second one derives a closed-form solution for per-antenna optimization, both being computationally efficient.

A. Conventional Phase-Only Beam Nulling Problem

Note that phase-only beam nulling can generate an independent beam with spatial interference suppression capability prior to the RF chain, thereby enabling the formation of interference-resistant beams equal in number to the available RF chains ($L = M$). Under the HAD architecture, phase-only beam nulling scheme presents only the analog beamformer which is responsible for both beam steering and interference suppression, while the digital beamformer does not perform additional spatial processing. Accordingly, the digital beamforming matrix is equivalent to an identity matrix $\mathbf{I}_{L \times L}$, i.e., it simply forward the signals from the RF chains without further weighting. This beamforming paradigm for the l th beam, $l = 1, \dots, L$, can be typically formulated as the following unit-modulus least squares (UMLSs) problem [42]:

$$P_4 : \min_{\mathbf{w}_l \in \mathbb{C}^{N_x \times N_y}} \|\mathbf{y} - \mathbf{A}_l^H \mathbf{w}_l\|_2^2 \quad (14a)$$

$$\text{s.t. } |[\mathbf{w}_l]_n| = 1 \quad \forall n \quad (14b)$$

where $\mathbf{y} \in \mathbb{C}^{K+1}$ denotes the desired array response. Herein, the first element of $\mathbf{y}$ should equal the main lobe gain to ensure accurate alignment of the l th beam toward (θ_B^l, ϕ_B^l) , while other elements should be chosen as small as possible to make sufficient reduction in (θ_I^k, ϕ_I^k) , $k = 1, \dots, K$. The

¹EIRP = $P_a \cdot G_a$ where P_a is the output power of power amplifiers and G_a is the antenna gain. In addition, the antenna gain is asymptotically given by $G_a \approx 4\pi A/\lambda^2$ where A is the antenna aperture area [63].

UMLS problem is NP-hard due to the unit-modulus constraint. Existing research solves this problem through semidefinite relaxation [40] or GP [42] at the cost of computational complexity that grows exponentially with array scale, which is particularly intolerable for LSAs. Based on the framework of majorization minimization, we have derived a closed-form solution and designed a bisection mechanism to accelerate convergence in our previous work [21], significantly lowering the computational complexity. However, since the desired array response $\mathbf{y}$ in P_4 is an array-specific parameter, it varies with the array configurations including the number of antenna elements and the array architectures, thereby limiting its generalizability across diverse array configurations.

B. Normalized Majorization Minimization

To ensure the generalizability of our approach without dependency on the array-specific parameter, we reformulate the beam nulling problem for the l th beam as the following normalized-UMLS problem:

$$P_5 : \min_{\mathbf{w}_l \in \mathbb{C}^{N_x N_y}, \eta_l \in \mathbb{C}} g_l(\mathbf{w}_l, \eta_l) = \|\mathbf{y} - \eta_l \mathbf{A}_l^H \mathbf{w}_l\|_2^2 \quad (15a)$$

$$\text{s.t. } \|\mathbf{w}_l\|_n = 1 \quad \forall n \quad (15b)$$

where η_l can be regarded as an automatic normalization factor to scale the achievable array response, and thus $\mathbf{y}$ can be set to the relative value, i.e., $\mathbf{y} = [1, 0, \dots, 0]^T \in \mathbb{C}^{K+1}$ for beam nulling. We address the additional variable in P_5 through alternating optimization. With a given $\mathbf{w}_l$, we have the derivation with respect to η_l as

$$\frac{\partial g_l(\mathbf{w}_l, \eta_l)}{\partial \eta_l} = \mathbf{w}_l^H \mathbf{A}_l \mathbf{A}_l^H \mathbf{w}_l \cdot \eta_l - \mathbf{w}_l^H \mathbf{A}_l \mathbf{y}. \quad (16)$$

By setting $\partial g_l(\mathbf{w}_l, \eta_l) / \partial \eta_l = 0$, the optimal scaling factor can be expressed as

$$\eta_l^* = \frac{\mathbf{w}_l^H \mathbf{A}_l \mathbf{y}}{\|\mathbf{A}_l^H \mathbf{w}_l\|_2^2}. \quad (17)$$

Subsequently, the objective function can be reformulated as

$$\begin{aligned} g_l(\mathbf{w}_l, \eta_l^*) &= (\mathbf{y} - \eta_l^* \mathbf{A}_l^H \mathbf{w}_l)^H (\mathbf{y} - \eta_l^* \mathbf{A}_l^H \mathbf{w}_l) \\ &= \zeta_l \cdot \mathbf{w}_l^H \mathbf{Q}_l \mathbf{w}_l - (\eta_l^* \mathbf{A}_l^H \mathbf{w}_l)^H \mathbf{y} - \mathbf{y}^H \eta_l^* \mathbf{A}_l^H \mathbf{w}_l + \mathbf{y}^H \mathbf{y} \\ &= \zeta_l \cdot \mathbf{w}_l^H \mathbf{Q}_l \mathbf{w}_l - 2\Re \left\{ (\eta_l^* \mathbf{A}_l^H \mathbf{w}_l)^H \mathbf{y} \right\} + \mathbf{y}^H \mathbf{y} \end{aligned} \quad (18)$$

where $\zeta_l = \eta_l^* (\eta_l^*)^H$ and $\mathbf{Q}_l = \mathbf{A}_l \mathbf{A}_l^H$. The quadratic term of (18) satisfies the following inequality [67]:

$$\begin{aligned} \mathbf{w}_l^H \mathbf{Q}_l \mathbf{w}_l &\leq 2\Re \left\{ \mathbf{w}_l^H (\mathbf{Q}_l - \tau_l \mathbf{I}) \right\} + \tau_l \mathbf{w}_l^H \mathbf{w}_l \\ &\quad + \mathbf{w}_{l,i-1}^H (\tau_l \mathbf{I} - \mathbf{Q}_l) \mathbf{w}_{l,i-1} \end{aligned} \quad (19)$$

where $\tau_l \mathbf{I} \geq \mathbf{Q}_l$ and $\mathbf{w}_{l,i-1}$ is a feasible solution to P_5 . Inspired by this inequality, through simple operations, we can construct the following upper-bounded function of the original function after substituting (19) to (18)

$$\begin{aligned} h_{l,i}(\mathbf{w}_l, \eta_l^*) &= 2\zeta_l \cdot \Re \left\{ \mathbf{w}_l^H [(\mathbf{Q}_l - \tau_l \mathbf{I}) \mathbf{w}_{l,i-1} - \mathbf{A}_l \mathbf{y} / \eta_l^*] \right\} + c_{l,i} \end{aligned} \quad (20)$$

where $c_{l,i} = 2\zeta_l \tau_l N_x N_y - \zeta_l \mathbf{w}_{l,i-1}^H \mathbf{Q}_l \mathbf{w}_{l,i-1} + \mathbf{y}^H \mathbf{y}$. Based on this majorizer function, we can formulate the following upper bound minimization problem:

$$P_6 : \min_{\mathbf{w}_l \in \mathbb{C}^{N_x N_y}} h_{l,i}(\mathbf{w}_l, \eta_l^*) \quad (21a)$$

$$\text{s.t. } \|\mathbf{w}_l\|_n = 1 \quad \forall n. \quad (21b)$$

Subsequently, we denote $\mathbf{w}_{l,i}$ as the optimal solution output by P_6 in the i th iteration. Since the inequality $g_l(\mathbf{w}_{l,i-1}, \eta_l^*) = h_{l,i}(\mathbf{w}_{l,i-1}, \eta_l^*) \geq h_{l,i}(\mathbf{w}_{l,i}, \eta_l^*) \geq g_l(\mathbf{w}_{l,i}, \eta_l^*)$ holds, we can solve P_5 through solving P_6 in a sequential manner, and the solution sequence $\{\mathbf{w}_{l,i}\}$ will keep the value of the original function $g_l(\mathbf{w}_{l,i}, \eta_l^*)$ monotonically decreasing as the number of iterations i increases. Additionally, with the scaling factor, the closed-form solution to P_6 can be simply derived according to the linear inner part of $\Re\{\cdot\}$ as well as the unit-modulus constraints, and it can be directly given by

$$\mathbf{w}_{l,i} = e^{j \cdot \arg[(\tau_l \mathbf{I} - \mathbf{Q}_l) \mathbf{w}_{l,i-1} + \mathbf{A}_l \mathbf{y} / \eta_l^*]}. \quad (22)$$

Algorithm 1 N-MM Algorithm for Beam Nulling

Input: $N_x, N_y, \mathbf{A}_l, \mathbf{y}, \mathbf{w}_{l,0} = e^{j \cdot \arg[(\mathbf{A}_l^H)^T \mathbf{y}]}, \epsilon \in \mathbb{R}^+$
Output: $\mathbf{w}_{l,i}$
 $\mathbf{Q}_l = \mathbf{A}_l \mathbf{A}_l^H, \tau_l = \text{tr}(\mathbf{Q}_l)$
for $i = 1, 2, \dots$ **do**
 $\tau_{l,i} = \tau_l / 2$
 $\eta_{l,i} = \mathbf{w}_{l,i-1}^H \mathbf{A}_l \mathbf{y} / \|\mathbf{A}_l^H \mathbf{w}_{l,i-1}\|_2^2$
 $\zeta_{l,i} = \eta_{l,i} \eta_{l,i}^H$
 for $t = 1$ **to** T **do**
 $c_{l,i} = 2\zeta_{l,i} \tau_{l,i} N_x N_y - \zeta_{l,i} \mathbf{w}_{l,i-1}^H \mathbf{Q}_l \mathbf{w}_{l,i-1} + \mathbf{y}^H \mathbf{y}$
 $\mathbf{w}_{l,i} = e^{j \cdot \arg[(\tau_{l,i} \mathbf{I} - \mathbf{Q}_l) \mathbf{w}_{l,i-1} + \mathbf{A}_l \mathbf{y} / \eta_{l,i}]}$
 $g_l(\mathbf{w}_{l,i}, \eta_{l,i}) = \|\mathbf{y} - \mathbf{A}_l^H \mathbf{w}_{l,i}\|_2^2$
 $h_{l,i}(\mathbf{w}_{l,i}, \eta_{l,i}) = 2\zeta_{l,i} \cdot$
 $\Re \left(\mathbf{w}_{l,i}^H [(\mathbf{Q}_l - \tau_{l,i} \mathbf{I}) \mathbf{w}_{l,i-1} - \mathbf{A}_l \mathbf{y} / \eta_{l,i}] \right) + c_{l,i}$
 if $g_l(\mathbf{w}_{l,i}, \eta_{l,i}) \leq h_{l,i}(\mathbf{w}_{l,i}, \eta_{l,i})$ **then**
 $\tau_{l,i} = \tau_{l,i} / 2$
 else
 $\tau_{l,i} = \tau_{l,i} + \tau_{l,i} / 2$
 end
 end
 if $g_l(\mathbf{w}_{l,i}, \eta_{l,i}) \leq \epsilon$, **then break**
end

Furthermore, according to [21, Lemma 3], a small enough value of τ_l can make sufficient reduction in $g_l(\mathbf{w}_l, \eta_l^*)$, thereby accelerating the convergence speed. Although $\tau_l = \text{tr}(\mathbf{Q}_l)$ exceeds the maximum eigenvalue and satisfies the above inequality, it is too large to facilitate rapid convergence. Therefore, we continue to utilize a simple yet effective bisection mechanism to search for an appropriate value of τ_l from the range of $[0, \text{tr}(\mathbf{Q}_l)]$. The overall normalized majorization minimization (N-MM) approach for phase-only beam nulling is summarized in Algorithm 1 where t is the iteration index of the bisection search and T denotes the maximum number of bisection iterations. Empirically, $T = 10$ bisections are enough to search for an appropriate τ_l . Additionally, the involved matrix $\mathbf{Q}_l$ exhibits a block Toeplitz structure, which can be

exploited to further reduce the complexity of matrix-vector multiplications [68]. It should be noted that the normalization factor is incorporated solely to resolve the beam pattern scaling ambiguity. Under the adopted alternating optimization framework with respect to η_l and $\mathbf{w}_l$, the following inequality chain consistently holds during the iterative process:

$$\begin{aligned} g_l(\mathbf{w}_{l,i-1}, \eta_{l,i-1}) &= h_{l,i}(\mathbf{w}_{l,i-1}, \eta_{l,i-1}) \geq h_{l,i}(\mathbf{w}_{l,i}, \eta_{l,i-1}) \\ &\geq g_l(\mathbf{w}_{l,i}, \eta_{l,i-1}) \geq g_l(\mathbf{w}_{l,i}, \eta_{l,i}) \end{aligned} \quad (23)$$

which indicates that, even with the introduction of the normalization factor, the value of the original objective function is monotonically nonincreasing over iterations until convergence is achieved. Repeating this algorithm with different manifold matrix $\mathbf{A}_l$, $l = 1, \dots, L$, we can obtain the entire analog beamforming matrix $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_L]$ for L beams. To exploit the potential of beam utilization, L can equal M to activate the maximum number of beams in this phase-only beam nulling manner.

C. Normalized Per-Antenna Optimization

In the above section, we present a normalized phase-only beam nulling approach based on majorization minimization which deals with the beamforming coefficients of the entire array in each iteration. From another perspective, we propose a per-antenna optimization approach in this section, where the beamforming coefficient for each antenna element is sequentially calculated with a closed-form solution.

Followed by the normalized-UMLS problem P_5 and the optimal normalized factor derived in (17), for a given $[\mathbf{w}_l]_{-n} \triangleq \mathbf{w}_l \setminus [\mathbf{w}_l]_n \in \mathbb{C}^{N_x N_y - 1}$ which denotes $\mathbf{w}_l$ with the n th element $[\mathbf{w}_l]_n$ removed, the phase-only beamforming problem for the n th element can be reformulated as

$$P_7: \min_{[\mathbf{w}_l]_n} \|\mathbf{y} - \eta_l^* \mathbf{A}_l^H \mathbf{w}_l\|_2^2 \quad \text{s.t. } |[\mathbf{w}_l]_n| = 1. \quad (24)$$

Expanding the above expression, the objective function of P_7 can be readily expressed as

$$\sum_k \left| [\mathbf{y}]_k - \eta_l^* [\mathbf{A}_l^H]_{k,n} [\mathbf{w}_l]_n - \sum_{n' \neq n} \eta_l^* [\mathbf{A}_l^H]_{k,n'} [\mathbf{w}_l]_{n'} \right|^2. \quad (25)$$

Similar to (18), by further expanding (25) and exploiting the property of conjugate summation, the expression can be decomposed into two parts: one that depends exclusively on $[\mathbf{w}_l]_n$, and the other that is independent of it. Specifically, the dependent part is given by

$$\begin{aligned} \sum_k 2\Re \left(\left| \eta_l^* \right|^2 [\mathbf{A}_l^H]_{k,n} [\mathbf{w}_l]_n \sum_{n' \neq n} [\mathbf{A}_l^H]_{k,n'}^* [\mathbf{w}_l]_{n'}^* \right. \\ \left. - \eta_l^* [\mathbf{y}]_k^* [\mathbf{A}_l^H]_{k,n} [\mathbf{w}_l]_n \right). \end{aligned} \quad (26)$$

Since P_7 is concerned exclusively with $[\mathbf{w}_l]_n$, the independent part can be ignored, and (26) can be directly adopted as the objective function. With this newly derived objective function, the original per-antenna optimization problem P_7 is equivalent to P_8 as in (27), shown at the bottom of the page.

Unlike the majorization minimization-based manner shown in the previous section, the per-antenna optimization problem P_8 requires solving for only a single variable at each iteration. Considering the phase-only constraint, we can explicitly represent $[\mathbf{w}_l]_n$ as $e^{j \cdot \arg([\mathbf{w}_l]_n)}$. With the real-part operator of a complex value, the optimal solution to P_8 should satisfy

$$[\mathbf{w}_l]_n^* \cdot e^{j \cdot \arg([\mathbf{w}_l]_n)} = -1 \quad (28)$$

where $\xi_{l,n}$ is given by

$$\xi_{l,n} = \sum_k \eta_l^* [\mathbf{A}_l^H]_{k,n} \left(\sum_{n' \neq n} (\eta_l^*)^* [\mathbf{A}_l^H]_{k,n'}^* [\mathbf{w}_l]_{n'}^* - [\mathbf{y}]_k^* \right). \quad (29)$$

Algorithm 2 N-PAO Algorithm for Beam Nulling

Input: $N_x, N_y, \mathbf{A}_l, \mathbf{y}, \mathbf{w}_{l,0} = e^{j \cdot \arg((\mathbf{A}_l^H)^\dagger \mathbf{y})}$, $\epsilon \in \mathbb{R}^+$
Output: $\mathbf{w}_{l,i}$
for $i = 1, 2, \dots$ **do**
 for $n = 1$ to $N_x N_y$ **do**
 $\eta_{l,i} = \mathbf{w}_{l,i-1}^H \mathbf{A}_l \mathbf{y} / \|\mathbf{A}_l^H \mathbf{w}_{l,i-1}\|^2$
 $\xi_{l,n} = \sum_k \eta_{l,i} [\mathbf{A}_l^H]_{k,n} \left(\sum_{n' \neq n} \eta_{l,i}^* [\mathbf{A}_l^H]_{k,n'}^* [\mathbf{w}_{l,i-1}]_{n'}^* - [\mathbf{y}]_k^* \right)$
 $[\mathbf{w}_{l,i-1}]_n = e^{j \cdot \arg(-\xi_{l,n}^*)}$
 end
 $\mathbf{w}_{l,i} = \mathbf{w}_{l,i-1}$
 If $g_l(\mathbf{w}_{l,i}, \eta_{l,i}) \leq \epsilon$, then break
end

Followed by (28), the optimal solution to P_8 can be expressed as $[\mathbf{w}_l]_n^* = e^{j \cdot \arg(-\xi_{l,n}^*)}$. After calculating the n th element, the per-antenna optimization continues to the next update until $N_x N_y$ elements have been calculated completely. Furthermore, we incorporate an iterative mechanism to ensure the objective function decreases progressively. Empirically, 10 iterations are sufficient for the algorithm to converge. The normalized per-antenna optimization (N-PAO) algorithm for phase-only beam nulling is summarized in Algorithm 2.

V. SIMULATION AND EXPERIMENTAL RESULTS

In this section, we provide a comprehensive evaluation of the proposed beam nulling techniques through both numerical simulations and experimental measurements. First, we quantitatively verify our analysis of existing HBF paradigms when applied to beam nulling, and evaluate the performance of the two proposed normalized phase-only beam nulling approaches through numerical simulations. Subsequently, we

$$P_8: \min_{[\mathbf{w}_l]_n} \Re \left([\mathbf{w}_l]_n \sum_k \eta_l^* [\mathbf{A}_l^H]_{k,n} \left(\sum_{n' \neq n} (\eta_l^*)^* [\mathbf{A}_l^H]_{k,n'}^* [\mathbf{w}_l]_{n'}^* - [\mathbf{y}]_k^* \right) \right) \quad \text{s.t. } |[\mathbf{w}_l]_n| = 1 \quad (27)$$

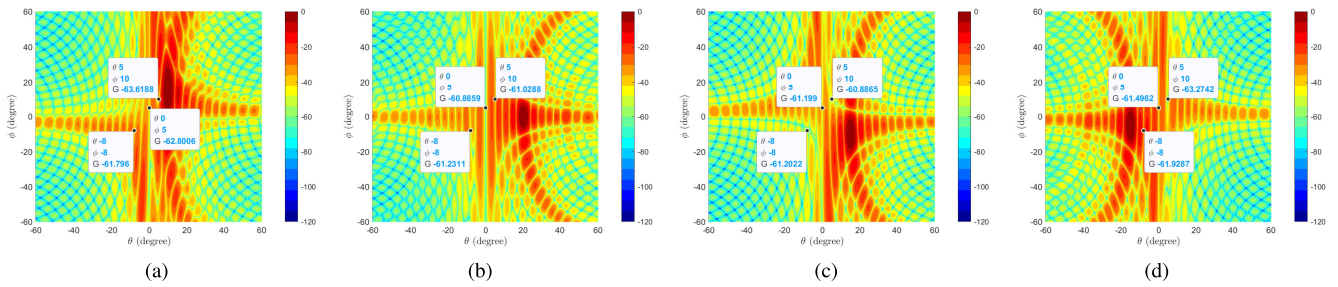


Fig. 4. Normalized beam patterns of the proposed N-MM algorithm where $M = L = 4$ and $K = 3$. (a) First beam pattern. (b) Second beam pattern. (c) Third beam pattern. (d) Fourth beam pattern.

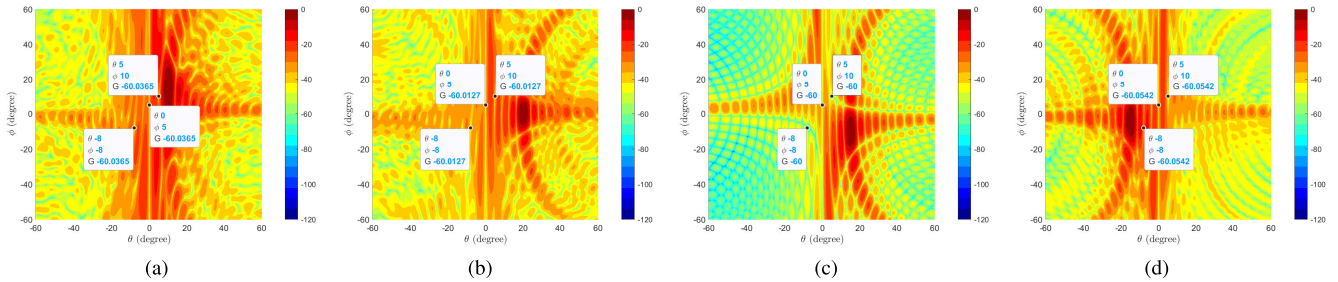


Fig. 5. Normalized beam patterns of the proposed N-PAO algorithm where $M = L = 4$ and $K = 3$. (a) First beam pattern. (b) Second beam pattern. (c) Third beam pattern. (d) Fourth beam pattern.

TABLE I
SIMULATION PARAMETERS

Parameters	Values
Satellite altitude	1000 km
Number of antenna elements	1024
Number of RF chains	4
System bandwidth	20 MHz
Center frequency	30 GHz
Noise temperature	500 K
Antenna aperture of users	0.25 m
Transmitted power of users	5 watts
Effective isotropic radiated power of users	42.67 dBW
Pointing direction of the first beam	$(10^\circ, 15^\circ)$
Pointing direction of the second beam	$(20^\circ, 0^\circ)$
Pointing direction of the third beam	$(15^\circ, -10^\circ)$
Pointing direction of the fourth beam	$(-15^\circ, -5^\circ)$
Antenna aperture of jammers	3 m
Transmitted power of jammers	1000 watts
Effective isotropic radiated power of jammers	87.26 dBW
Arriving direction of the first jammer	$(0^\circ, 5^\circ)$
Arriving direction of the second jammer	$(-8^\circ, -8^\circ)$
Arriving direction of the third jammer	$(5^\circ, 10^\circ)$

TABLE II
COMPARISON OF BEAM NULLING PERFORMANCE (dB)

Algorithms		Jammer 1	Jammer 2	Jammer 3
N-MM	Beam 1	-62.80	-61.79	-63.62
	Beam 2	-61.03	-61.23	-60.89
	Beam 3	-60.89	-61.20	-61.20
	Beam 4	-63.27	-61.93	-61.50
N-PAO	Beam 1	-60.03	-60.03	-60.03
	Beam 2	-60.01	-60.01	-60.01
	Beam 3	-60.00	-60.00	-60.00
	Beam 4	-60.05	-60.05	-60.05

further demonstrate the practical viability of the proposed methods using a proof-of-concept phased array prototype, thereby validating their practical applicability in interference mitigation under realistic hardware constraints.

A. Beam Pattern Characteristics

Consider a satellite communication system equipped with a 32×32 uniform rectangular HAD array, in which all antenna elements are connected to 4 RF chains, i.e., $N_x = N_y = 32$ and $M = 4$. To demonstrate the beam utilization of phase-only beam nulling, we consider all RF chains are leveraged to form independent beams, i.e., $L = M = 4$. Suppose that all four

beams are required to suppress three ground-based interference sources, with the pointing directions and interference directions listed in Table I. To mitigate interference, the desired normalized reception gain of interference is set to 10^{-6} which is already sufficient to reduce the interference power below the noise level. With the unit-modulus beamforming matrix output by our proposed N-MM and N-PAO algorithms, the 2-D beam pattern for each beam is illustrated in Figs. 4 and 5 respectively. As shown in both figures, the main lobe of each beam is accurately steered toward the desired direction while three deep nulls are placed toward the three jammer directions. To quantitatively evaluate the interference suppression performance, Table II reports the achieved normalized gains at the jammer directions for all beams. It can be observed that both algorithms consistently achieve deep nulls below -60 dB for all beams and all interference sources. Specifically, the N-MM algorithm attains null depths ranging approximately from -63.6 to -60.9 dB, while the N-PAO algorithm yields null depths tightly clustered around -60 dB.

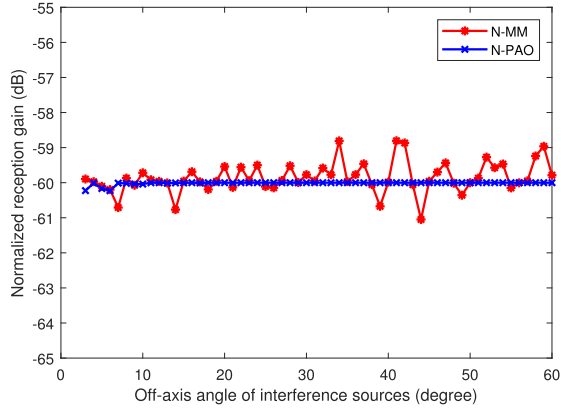


Fig. 6. Normalized reception gain versus the locations of interference sources.

TABLE III
INPUT POWER OF EACH RF CHAIN (dBm)

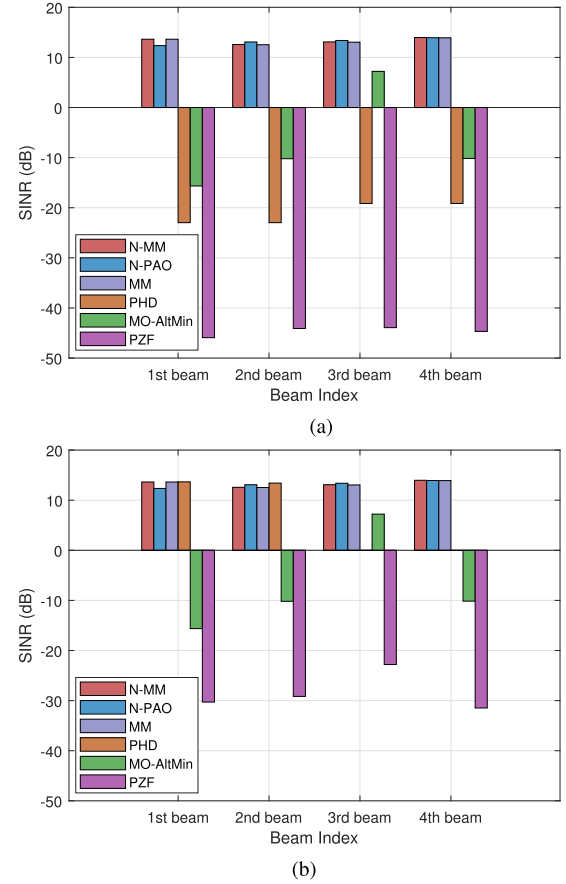
Algorithms	Channel #1	Channel #2	Channel #3	Channel #4
N-MM	-34.93	-34.80	-34.78	-34.79
N-PAO	-35.92	-35.24	-34.86	-35.26
MM	-34.90	-34.75	-34.68	-34.73
PHD	4.29	4.29	0.60	0.60
MO-AltMin	-2.85	-8.27	-26.27	-8.33
PZF	18.77	19.08	18.46	18.64

A notable difference between the two algorithms lies in the stability of the null depths. The N-PAO method produces nearly identical suppression levels across different beams and jammers, indicating a highly uniform nulling behavior. In contrast, the N-MM method exhibits small fluctuations in null depth across beams. On the other hand, when the locations of the interference sources change, Fig. 6 indicates that both N-MM and N-PAO algorithms can consistently control the reception gains at significantly lower levels in the corresponding directions, albeit N-MM exhibits fluctuation within a 1 dB range. The beam pattern characteristics validate the effectiveness of the two proposed normalized phase-only beam nulling approaches, which simultaneously exploit the available DoF of HAD arrays and suppress malicious interference.

B. SINR Performance

We further compare the SINR performance of different HBF paradigms, where the PHD method for perfect analog-digital decomposition [33], the MO-AltMin method for SU-HBF [28], the PZF method for MU-HBF [32], and our previously proposed MM method for nonnormalized phase-only beam nulling [21] are presented for comparison. Herein, the adopted simulation parameters, including satellite altitude, operating frequency, and EIRP, are listed in Table I. Through link budget calculation, the SINR^2 performance of different HBF paradigms is plotted in Fig. 7, where interbeam interference and external interference are simultaneously considered. When dealing with strong malicious interference, it is discovered

²Generally, the noise power can be calculated by $\sigma_n^2 = \kappa TB$ where κ is the Boltzmann's constant, T is the noise temperature in Kelvins, and B is the system bandwidth.

Fig. 7. SINR versus different HBF paradigms. (a) Only with the analog matrix W . (b) With the analog-digital matrix WF .

from Fig. 7(a) that the typical HBF paradigms, especially the PHD and PZF methods, output extremely low SINR at the RF front end when exclusively using the analog beamforming matrix W . Combined with the low-dimensional digital beamforming F , as shown in Fig. 7(b), the PHD method exhibits optimal DBF performance on only two beams, effectively filtering out interference and resulting in higher SINR. This behavior stems from the fact that, as shown in Proposition 1, sacrificing half of the beams allows the analog-digital decomposition to exactly reproduce the FD beamformer, enabling complete interference suppression in the digital stage. However, the SINR performance of the MO-AltMin and PZF methods still remains poor, since MO-AltMin suffers from unavoidable matrix fitting errors as established in Proposition 2, while PZF is fundamentally limited by the insufficient DoF. The numerical results in this section further demonstrate our analysis in Section III that the existing HBF paradigms will cause beam distortion which cannot effectively suppress interference, or need to reduce the number of activating beams for anti-interference capability. In contrast, all phase-only beam nulling methods, including the nonnormalized MM algorithm, the proposed N-MM and N-PAO algorithms, achieve similar SINR performance, with each beam exhibiting higher SINR on both analog and digital sides. This also indicates that the interference suppression performance remains unaffected by the introduction of a normalization factor. Additionally, with

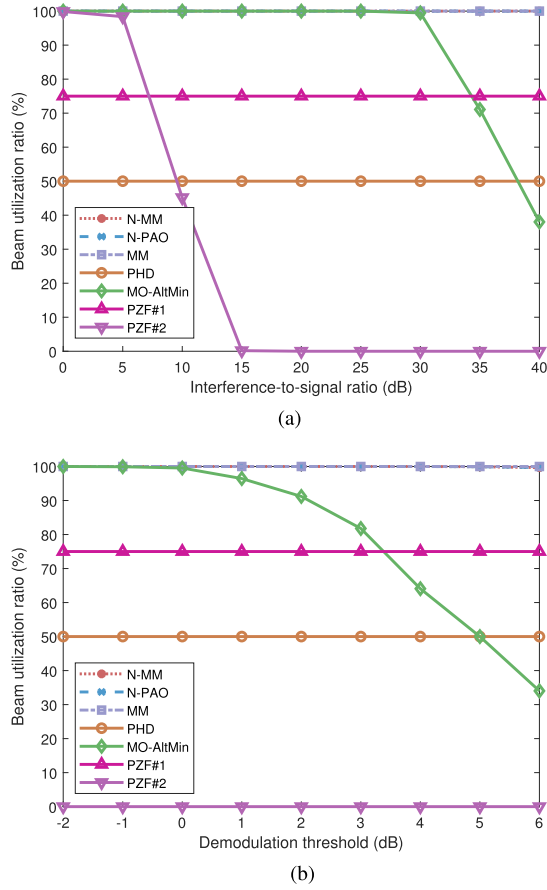


Fig. 8. Beam utilization ratio for different HBF paradigms. (a) Beam utilization ratio versus interference-to-signal ratio. (b) Beam utilization ratio versus demodulation threshold.

an assumed LNA gain of 25 dB, the input power of each RF chain after analog beamforming is listed in Table III. These results reveal that the phase-only beam nulling achieves significantly lower interference power, whereas the typical HBF paradigms could potentially lead to RF-chain saturation due to the significantly higher input power caused by the insufficient analog beamforming part exclusively. Overall, phase-only beam nulling emerges as a preferred interference mitigation strategy for multibeam HAD arrays, ensuring improved SINR performance and avoiding RF-chain saturation under powerful malicious jamming attacks.

C. Beam Utilization Efficiency

The above section considers the reception performance when encountering extremely powerful malicious jammers. Herein, we further evaluate the generalized multibeam capacity of satellite communication systems under different-level malicious jamming attacks, and the achievable beam utilization efficiency is defined as

$$\zeta = \mathbb{E} \left(\frac{\sum_{l=1}^L \delta(\gamma_l \geq \gamma_{th})}{M} \right) \quad (30)$$

where $\mathbb{E}(\cdot)$ denotes the expectation operator, γ_{th} denotes the demodulation threshold, $\delta(\cdot)$ denotes the indicator function which equals 1 if the inner condition is satisfied, or equals 0

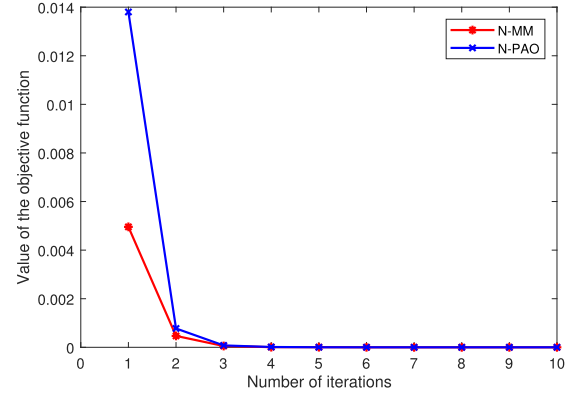


Fig. 9. Objective function versus the number of iterations.

otherwise. Note that the denominator of (30) is the number of RF chains (the maximum number of achievable beams, i.e., $L \leq M$), and the numerator is the actual number of available beams whose received SINR exceeds the demodulation threshold. To facilitate analysis, we consider a single malicious interference source randomly located in a different direction in each Monte Carlo experiment, and 1000 Monte Carlo trials are executed to calculate the beam utilization efficiency which is plotted in Fig. 8. The notation PZF#1 represents the case where the sufficient DoF is achieved for interference suppression at the cost of sacrificing one beam through the PZF method, while PZF#2 denotes the case of the insufficient DoF instead. In Fig. 8(a) where the demodulation threshold γ_{th} is set to 0 dB, the phase-only beam nulling methods, including N-MM, N-PAO and MM, can consistently achieve 100% beam utilization across interference attacks of varying intensities. In contrast, the beam utilization of the PZF#1 and PHD methods reach only 75% and 50% respectively, indicating that only 3 beams and 2 beams are formed despite the availability of 4 RF chains. Once the number of interference sources increases, the PZF method exhibits lower beam utilization efficiency since more beams must be traded off for anti-interference capability. Regarding the PZF#2 and MO-AltMin method, due to beam distortion as illustrated in Fig. 2, the received SINR is tolerable when the interference power is low, whereas the beam utilization is significantly degraded under powerful interference conditions. On the other hand, as the demodulation threshold changes, similar phenomena can be observed in Fig. 8(b) where the input interference-to-signal ratio is set to 30 dB. In summary, comparative analysis reveals that the phase-only beam nulling paradigm exhibits both high multibeam capability as well as superior interference suppression performance.

D. Computational Complexity and Convergence

Herein, we first evaluate the computational complexity of the two proposed normalized beam nulling algorithms, where the execution times of existing phase-only approaches, including our previous work (MM) [21], the GP method [42], and the semidefinite (SDR) method [40], are presented for comparison. Herein, all algorithms are executed on an i7-10700K CPU with 64GB RAM (operating at 4 GHz). We conducted 1000

TABLE IV
EXECUTION TIME OF DIFFERENT ALGORITHMS

Algorithm	Computational complexity	Execution time
N-MM	$\mathcal{O}(I_1 T_1 N_x N_y \log_2(N_x N_y) + \epsilon_s)$	0.121 s
N-PAO	$\mathcal{O}(I_2 (N_x N_y)^2 + \epsilon_s)$	0.654 s
MM	$\mathcal{O}(I_3 T_2 N_x N_y \log_2(N_x N_y))$	0.085 s
GP	$\mathcal{O}(I_4 (N_x N_y)^3)$	26.013 s
SDR	$\mathcal{O}((N_x N_y)^7)$	–

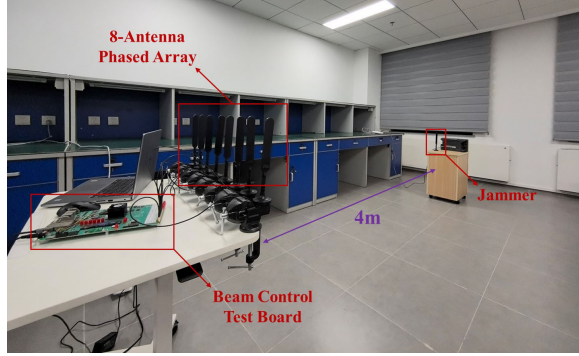
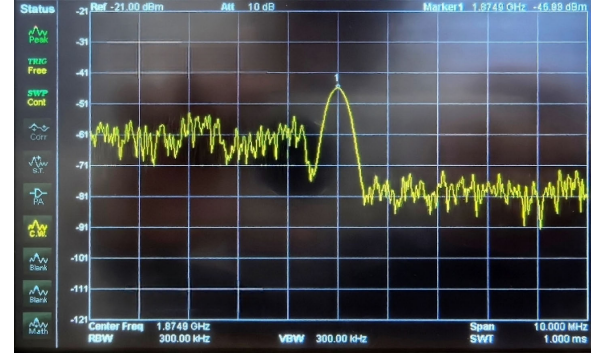


Fig. 10. Experiment setup.

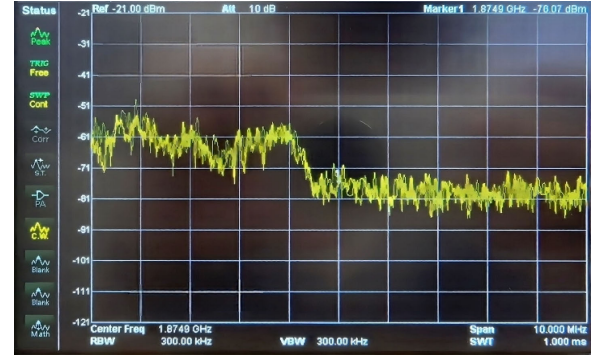
Monte-Carlo trials to measure the execution time of each algorithm, and the results are presented in Table IV where $I_p, p \in \{1, 2, 3, 4\}$ denotes the number of iterations. The SDR method fails in this case because the required RAM for 1024 antennas exceeds the CPU configuration, which demonstrates its inadequacy for large-scale arrays. Since the two proposed methods and our previous work are exclusively implemented through matrix-vector multiplications and additions, the execution times of them are all significantly lower than that of the GP method which requires eigenvalue decomposition of large-scale matrices. Compared to our previous work, the additional term of the proposed normalized approaches is to calculate the normalization factor, which adds only marginal computational complexity. Consequently, the total execution times of the two proposed approaches are slightly higher than that of our previous work. However, it should be noted that the normalized beam nulling methods N-MM and N-PAO exhibit superior generalizability across different array configurations, since the requirement for parameter adjustment which depends on array configurations is eliminated. Furthermore, Fig. 9 illustrates the evolution of the objective function value versus the number of iterations for the proposed N-MM and N-PAO algorithms. Specifically, the objective values of both algorithms drop sharply during the first few iterations and approach zero after approximately three to four iterations, indicating fast convergence. The results in Fig. 9 confirm that the introduction of the automatic normalization factor does not hinder convergence and allows both algorithms to maintain stable and efficient iterative behavior.

E. Prototype Evaluation

In a laboratory environment, we develop an eight-antenna phased array prototype, with the phase-only beam nulling



(a)



(b)

Fig. 11. Spectral measurement results of the prototype. (a) Spectrum before phase-only beam nulling. (b) Spectrum after phase-only beam nulling.

realized on a specially designed beam control test board to validate the proposed beam nulling algorithms. Under laboratory conditions, the amplitude–phase errors of the antenna array prototype are precompensated, and mutual coupling effects are alleviated by increasing the interelement spacing. However, in practical satellite-borne system deployments, such impairments may need to be explicitly mitigated, which constitutes an important direction for our future research. The receiving array comprises omnidirectional radiating elements, each connected to the beam control board which enables independent phase adjustment across different antenna channels. To accommodate the hardware constraints inherent in practical implementations, specifically the finite resolution of the PSs employed in this experimental setup, all phase control values are quantized to 6-bit precision using the following minimum-error quantization method [33], [69]

$$\min_{\hat{\mathbf{w}}_l} \|\mathbf{w}_l - \hat{\mathbf{w}}_l\|_2^2 \quad \text{s.t.} \quad [\hat{\mathbf{w}}_l]_n \in \left\{1, e^{j\frac{2\pi}{2^6}}, \dots, e^{j(2^6-1)\frac{2\pi}{2^6}}\right\}. \quad (31)$$

Due to the limitations of the available experimental facilities, our evaluation focuses exclusively on characterizing the interference suppression performance by measuring the received interference power variation before and after applying the beam nulling technique. As shown in Fig. 10, the experimental configuration consists of a jammer positioned 4 m away from the receiving array, which continuously transmits a single-tone interference signal through a single omnidirectional antenna. Prior to beam nulling, the spectrum of the interference signal is captured in Fig. 11(a), where the measured interference power is -46.08 dBm, significantly above the noise floor. After

applying either the proposed N-MM or N-PAO algorithm, the interference is dramatically suppressed, with the residual power reduced below the noise level as depicted in Fig. 11(b). This substantial power reduction validates the practical efficacy of both algorithms, and confirms that despite 6-bit phase quantization, both methods achieve effective null formation under hardware imperfections.

VI. CONCLUSION

In this article, considering the multibeam satellite communication systems equipped with HAD arrays, we analyze the performance of existing HBF paradigms when applied to beam nulling for interference mitigation. Through comparative analysis, we reveal that phase-only beam nulling emerges as the preferable interference mitigation paradigm for HAD arrays, since it can effectively suppress interference while fully exploiting multibeam capacity, with the additional advantage to avoid RF chain saturation caused by the extremely powerful malicious interference. Furthermore, we propose two normalized phase-only beam nulling approaches which introduce an automatic normalization factor to scale the achievable beam pattern, thereby eliminating the requirement for array-dependent parameters. Specifically, the N-MM method designs a simple bisection mechanism to accelerate convergence, while the N-PAO approach reduces computational complexity through a closed-form solution for each antenna element. Numerical simulations confirm our analysis of interference mitigation paradigms, and demonstrate that the proposed phase-only beam nulling approaches can effectively suppress interference at significantly lower computational complexity. Experimental measurements with a phased array prototype further validate the practical effectiveness of both methods under realistic hardware constraints. It is worth noting that beam nulling becomes highly challenging when the number of jammers approaches the array size. While this work considers typical satellite scenarios with limited jammers relative to the array size, future research will investigate null broadening techniques for dense jamming scenarios approaching the spatial DoF limit.

APPENDIX A

PROOF OF PROPOSITION 2

Without loss of generality, we consider a single interference source in this proof, but the analysis can be readily extended to the multisource scenario. The existing research has demonstrated that $\mathbf{V} \neq \mathbf{W}^* \mathbf{F}^*$ in the case where $M/2 < L \leq M$. Consequently, there must exist $l' \in \{1, \dots, L\}$ such that the overall HBF vector $\mathbf{W}^* \mathbf{f}_l^*$ for the l' th beam results in $\mathbf{W}^* \mathbf{f}_l^* = \mathbf{v}_{l'} + \boldsymbol{\epsilon}_{l'}$ where $\boldsymbol{\epsilon}_{l'} \neq \mathbf{0}$. By substituting this formula, we can reformulated the overall HAD array gain in the direction of interference as

$$\begin{aligned} & |\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{W}^* \mathbf{f}_l^*|^2 \\ &= |\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{v}_{l'}|^2 + |\mathbf{a}^H(\theta_l^k, \phi_l^k) \boldsymbol{\epsilon}_{l'}|^2 \\ &+ 2\Re\{\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{v}_{l'} \boldsymbol{\epsilon}_{l'}^H \mathbf{a}(\theta_l^k, \phi_l^k)\} \end{aligned} \quad (32)$$

where $\Re\{\cdot\}$ represents the real part operator, and the first summation term on the right side is the corresponding FD

array gain. For the third summation term on the right side, we have the following inequality

$$\begin{aligned} & 2\Re\{\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{v}_{l'} \boldsymbol{\epsilon}_{l'}^H \mathbf{a}(\theta_l^k, \phi_l^k)\} \\ & \geq -2|\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{v}_{l'}| |\mathbf{a}^H(\theta_l^k, \phi_l^k) \boldsymbol{\epsilon}_{l'}|. \end{aligned} \quad (33)$$

For simplification, we define that $\rho_H \triangleq |\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{W}^* \mathbf{f}_l^*|$, $\rho_D \triangleq |\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{v}_{l'}|$ and $\rho_E \triangleq |\mathbf{a}^H(\theta_l^k, \phi_l^k) \boldsymbol{\epsilon}_{l'}|$, followed by the inequality $\rho_H^2 \geq \rho_D^2 + \rho_E^2 - 2\rho_D\rho_E$. According to the [70, Lemma 2], for any given (θ_B^l, ϕ_B^l) and (θ_l^k, ϕ_l^k) satisfying $\mathbf{a}(\theta_B^l, \phi_B^l) \neq \beta \mathbf{a}(\theta_l^k, \phi_l^k)$ where $\forall \beta \in \mathbb{C}$, then for $\forall \rho_D > 0$, there must exist $\mu \in \mathbb{C}$ and $\mathbf{v}_{l'} = \mathbf{a}(\theta_B^l, \phi_B^l) + \mu \mathbf{a}(\theta_l^k, \phi_l^k)$ such that $|\mathbf{a}^H(\theta_l^k, \phi_l^k) \mathbf{v}_{l'}| = \rho_D$. Hence, when the interference source is located in the sidelobe region, we can determine the parameter μ to construct the DBF vector $\mathbf{v}_{l'}$ which satisfies $\rho_D \in (0, \rho_E/2)$ and serves as a feasible solution to P_1 simultaneously. In this regard, we have the inequality $\rho_H^2 \geq \rho_D^2 + \rho_E^2 - 2\rho_D\rho_E > \rho_D^2$, and it is verified that the solution to P_2 will lead to higher HAD array gain in the direction of interference compared to that of the FD array, which completes the proof.

REFERENCES

- [1] J. Lin, W. Yu, N. Zhang, X. Yang, H. Zhang, and W. Zhao, "A survey on Internet of Things: Architecture, enabling technologies, security and privacy, and applications," *IEEE Internet Things J.*, vol. 4, no. 5, pp. 1125–1142, Oct. 2017.
- [2] J. Liu, Y. Shi, Z. M. Fadlullah, and N. Kato, "Space-air-ground integrated network: A survey," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2714–2741, 4th Quart., 2018.
- [3] L. Zhi et al., "Self-powered absorptive reconfigurable intelligent surfaces for securing satellite-terrestrial integrated networks," *China Commun.*, vol. 21, no. 9, pp. 276–291, Sep. 2024.
- [4] Y. Xiao et al., "Space-air-ground integrated wireless networks for 6G: Basics, key technologies, and future trends," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 12, pp. 3327–3354, Dec. 2024.
- [5] J. A. Fraire, O. Iova, and F. Valois, "Space-terrestrial integrated Internet of Things: Challenges and opportunities," *IEEE Commun. Mag.*, vol. 60, no. 12, pp. 64–70, Dec. 2022.
- [6] M. Vaezi et al., "Cellular, wide-area, and non-terrestrial IoT: A survey on 5G advances and the road toward 6G," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 1117–1174, 2nd Quart., 2022.
- [7] X. Zhong, B. Ren, X. Gong, and H. Li, "Traffic load optimization for multi-satellite relay systems in space information network: A proportional fairness approach," *Sensors*, vol. 22, no. 22, p. 8806, Nov. 2022.
- [8] Y. He, Y. Liu, C. Jiang, and X. Zhong, "Multiobjective anti-collision for massive access ranging in MF-TDMA satellite communication system," *IEEE Internet Things J.*, vol. 9, no. 16, pp. 14655–14666, Aug. 2022.
- [9] V. Jorroughi, M. A. Vázquez, A. I. Pérez-Neira, and B. Devillers, "Onboard beam generation for multibeam satellite systems," *IEEE Trans. Wireless Commun.*, vol. 16, no. 6, pp. 3714–3726, Jun. 2017.
- [10] P. J. Honnaiah, N. Maturo, S. Chatzinotas, S. Kisseleff, and J. Krause, "Demand-based adaptive multi-beam pattern and footprint planning for high throughput GEO satellite systems," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 1526–1540, 2021.
- [11] J. Wang, C. Jiang, L. Kuang, and R. Han, "Satellite multi-beam collaborative scheduling in satellite aviation communications," *IEEE Trans. Wireless Commun.*, vol. 23, no. 3, pp. 2097–2111, Mar. 2024.
- [12] Z. Lin, Z. Ni, L. Kuang, C. Jiang, and Z. Huang, "Dynamic beam pattern and bandwidth allocation based on multi-agent deep reinforcement learning for beam hopping satellite systems," *IEEE Trans. Veh. Technol.*, vol. 71, no. 4, pp. 3917–3930, Apr. 2022.
- [13] C. Han, A. Liu, L. Huo, H. Wang, X. Liang, and X. Tong, "Distributed resource management framework for IoS against malicious jamming," *IEEE Trans. Commun.*, vol. 69, no. 12, pp. 8271–8286, Dec. 2021.
- [14] Z. Xu, B. Tang, W. Ai, and J. Zhu, "Relative entropy based jamming signal design against radar target detection," *IEEE Trans. Signal Process.*, vol. 73, pp. 1200–1215, 2025.

- [15] C. Zou et al., "Multi-layer RIS-assisted anti-jamming communications: A hierarchical game learning approach," *IEEE Commun. Lett.*, vol. 27, no. 11, pp. 2998–3002, Nov. 2023.
- [16] H. Baek and J. Lim, "Spectrum sharing for coexistence of fixed satellite services and frequency hopping tactical data link," *IEEE J. Sel. Areas Commun.*, vol. 34, no. 10, pp. 2642–2649, Oct. 2016.
- [17] Y. Gao, Y. Xiao, M. Wu, M. Xiao, and J. Shao, "Game theory-based anti-jamming strategies for frequency hopping wireless communications," *IEEE Trans. Wireless Commun.*, vol. 17, no. 8, pp. 5314–5326, Aug. 2018.
- [18] Á. Vázquez-Castro, "Asymptotically guaranteed anti-jamming spread spectrum random access without pre-shared secret," *IEEE Trans. Inf. Forensics Security*, vol. 17, pp. 332–343, 2022.
- [19] H. Pirayesh and H. Zeng, "Jamming attacks and anti-jamming strategies in wireless networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 767–809, 2nd Quart., 2022.
- [20] Q. Wei, Y. Gao, X. Chen, Z. Huang, and L. Kuang, "Spatial filtering for interference mitigation in high-altitude spaceborne GNSS receivers," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 1, pp. 274–291, Feb. 2023.
- [21] Z. Chen, L. Kuang, B. Liu, and Z. Ni, "Fast and efficient phase-only beam nulling for NGSO satellite communication systems," *IEEE Wireless Commun. Lett.*, vol. 13, no. 12, pp. 3261–3265, Dec. 2024.
- [22] W. Hong et al., "Multibeam antenna technologies for 5G wireless communications," *IEEE Trans. Antennas Propag.*, vol. 65, no. 12, pp. 6231–6249, Dec. 2017.
- [23] I. del Portillo, B. G. Cameron, and E. F. Crawley, "A technical comparison of three low Earth orbit satellite constellation systems to provide global broadband," *Acta Astronautica*, vol. 159, pp. 123–135, Jun. 2019.
- [24] X. Gao, L. Dai, and A. M. Sayeed, "Low RF-complexity technologies to enable millimeter-wave MIMO with large antenna array for 5G wireless communications," *IEEE Commun. Mag.*, vol. 56, no. 4, pp. 211–217, Apr. 2018.
- [25] I. Ahmed et al., "A survey on hybrid beamforming techniques in 5G: Architecture and system model perspectives," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 3060–3097, 2018.
- [26] J. A. Zhang, X. Huang, V. Dyadyuk, and Y. J. Guo, "Massive hybrid antenna array for millimeter-wave cellular communications," *IEEE Wireless Commun.*, vol. 22, no. 1, pp. 79–87, Feb. 2015.
- [27] O. E. Ayach, S. Rajagopal, S. Abu-Surra, Z. Pi, and R. W. Heath Jr., "Spatially sparse precoding in millimeter wave MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 3, pp. 1499–1513, Mar. 2014.
- [28] X. Yu, J.-C. Shen, J. Zhang, and K. B. Letaief, "Alternating minimization algorithms for hybrid precoding in millimeter wave MIMO systems," *IEEE J. Sel. Topics Signal Process.*, vol. 10, no. 3, pp. 485–500, Apr. 2016.
- [29] T. Lin, J. Cong, Y. Zhu, J. Zhang, and K. Ben Letaief, "Hybrid beamforming for millimeter wave systems using the MMSE criterion," *IEEE Trans. Commun.*, vol. 67, no. 5, pp. 3693–3708, May 2019.
- [30] R. Mai, D. H. N. Nguyen, and T. Le-Ngoc, "MMSE hybrid precoder design for millimeter-wave massive MIMO systems," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Apr. 2016, pp. 1–6.
- [31] X. Zhao, T. Lin, Y. Zhu, and J. Zhang, "Partially-connected hybrid beamforming for spectral efficiency maximization via a weighted MMSE equivalence," *IEEE Trans. Wireless Commun.*, vol. 20, no. 12, pp. 8218–8232, Dec. 2021.
- [32] L. Liang, W. Xu, and X. Dong, "Low-complexity hybrid precoding in massive multiuser MIMO systems," *IEEE Wireless Commun. Lett.*, vol. 3, no. 6, pp. 653–656, Dec. 2014.
- [33] F. Sohrabi and W. Yu, "Hybrid digital and analog beamforming design for large-scale antenna arrays," *IEEE J. Sel. Topics Signal Process.*, vol. 10, no. 3, pp. 501–513, Apr. 2016.
- [34] Q. Shi and M. Hong, "Spectral efficiency optimization for millimeter wave multiuser MIMO systems," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 3, pp. 455–468, Jun. 2018.
- [35] M. Hui, X. Zhao, T. Lin, and Y. Zhu, "Hybrid beamforming for utility maximization in multiuser broadband millimeter wave systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 12, pp. 16042–16057, Dec. 2023.
- [36] A. M. Elbir, K. V. Mishra, S. A. Vorobyov, and R. W. Heath Jr., "Twenty-five years of advances in beamforming: From convex and nonconvex optimization to learning techniques," *IEEE Signal Process. Mag.*, vol. 40, no. 4, pp. 118–131, Jun. 2023.
- [37] R. L. Haupt, "Phase-only adaptive nulling with a genetic algorithm," *IEEE Trans. Antennas Propag.*, vol. 45, no. 6, pp. 1009–1015, Jun. 1997.
- [38] S. T. Smith, "Optimum phase-only adaptive nulling," *IEEE Trans. Signal Process.*, vol. 47, no. 7, pp. 1835–1843, Jul. 1999.
- [39] X. Jiang, J. Chen, X. Liu, A. M. Zoubir, and Z. Zhou, "Phase-only robust minimum dispersion beamforming," *IEEE Trans. Signal Process.*, vol. 68, pp. 5664–5679, 2020.
- [40] P. J. Kajenski, "Phase only antenna pattern notching via a semidefinite programming relaxation," *IEEE Trans. Antennas Propag.*, vol. 60, no. 5, pp. 2562–2565, May 2012.
- [41] Y. Lee, "Adaptive interference suppression of phase-only thinned arrays via convex optimization," *IEEE Trans. Antennas Propag.*, vol. 68, no. 6, pp. 4583–4592, Jun. 2020.
- [42] J. Tranter, N. D. Sidiropoulos, X. Fu, and A. Swami, "Fast unit-modulus least squares with applications in beamforming," *IEEE Trans. Signal Process.*, vol. 65, no. 11, pp. 2875–2887, Jun. 2017.
- [43] M. Zhang, J. Li, S. Zhu, and X. Chen, "Fast and simple gradient projection algorithms for phase-only beamforming," *IEEE Trans. Veh. Technol.*, vol. 70, no. 10, pp. 10620–10632, Oct. 2021.
- [44] X. Zhang, Z. He, B. Liao, and X. Zhang, "Fast array response adjustment with phase-only constraint: A geometric approach," *IEEE Trans. Antennas Propag.*, vol. 67, no. 10, pp. 6439–6451, Oct. 2019.
- [45] X. Huang, Y. J. Guo, and J. D. Bunton, "A hybrid adaptive antenna array," *IEEE Trans. Wireless Commun.*, vol. 9, no. 5, pp. 1770–1779, May 2010.
- [46] R. Zhang, B. Shim, and W. Wu, "Direction-of-arrival estimation for large antenna arrays with hybrid analog and digital architectures," *IEEE Trans. Signal Process.*, vol. 70, pp. 72–88, 2022.
- [47] Z. Chen, L. Kuang, Z. Ni, B. Liu, and Z. Lin, "Two-stage coprime matching angle-of-arrival estimation for hybrid analog and digital arrays," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 6050–6064, 2026.
- [48] T. Schenk, *RF Imperfections in High-Rate Wireless Systems: Impact and Digital Compensation*. Cham, Switzerland: Springer, 2008.
- [49] U. Gustavsson et al., "On the impact of hardware impairments on massive MIMO," in *Proc. IEEE GLOBECOM Workshops (GC Wkshps)*, Dec. 2014, pp. 294–300.
- [50] E. Björnson, J. Hoydis, M. Kountouris, and M. Debbah, "Massive MIMO systems with non-ideal hardware: Energy efficiency, estimation, and capacity limits," *IEEE Trans. Inf. Theory*, vol. 60, no. 11, pp. 7112–7139, Nov. 2014.
- [51] S. Gong et al., "Hardware-impaired RIS-assisted mmWave hybrid systems: Beamforming design and performance analysis," *IEEE Trans. Commun.*, vol. 71, no. 4, pp. 2317–2334, Apr. 2023.
- [52] H. Xu, Y. Sun, Y. Zhao, M. Peng, and S. Zhang, "Joint beam scheduling and beamforming design for cooperative positioning in multi-beam LEO satellite networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 4, pp. 5276–5287, Apr. 2024.
- [53] Z. Yin et al., "Multi-domain resource multiplexing based secure transmission for satellite-assisted IoT: AO-SCA approach," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 7319–7330, Nov. 2023.
- [54] Z. Yin, N. Cheng, T. H. Luan, Y. Hui, and W. Wang, "Green interference based symbiotic security in integrated satellite-terrestrial communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 11, pp. 9962–9973, Nov. 2022.
- [55] H. L. Van Trees, *Optimum Array Processing: Part IV of Detection, Estimation, and Modulation Theory*. Hoboken, NJ, USA: Wiley, 2002.
- [56] A. A. M. Saleh, "Frequency-independent and frequency-dependent non-linear models of TWT amplifiers," *IEEE Trans. Commun.*, vol. COM-29, no. 11, pp. 1715–1720, Nov. 1981.
- [57] J. K. Cavers, "Amplifier linearization using a digital predistorter with fast adaptation and low memory requirements," *IEEE Trans. Veh. Technol.*, vol. 39, no. 4, pp. 374–382, Apr. 1990.
- [58] Verspecht and D. E. Root, "Polyharmonic distortion modeling," *IEEE Microw. Mag.*, vol. 7, no. 3, pp. 44–57, Jun. 2006.
- [59] O. T. Demir and E. Björnson, "The Bussgang decomposition of non-linear systems: Basic theory and MIMO extensions [Lecture Notes]," *IEEE Signal Process. Mag.*, vol. 38, no. 1, pp. 131–136, Jan. 2021.
- [60] Y. Li, J. Cervantes, N. C. Shivaramaiah, D. M. Akos, and M. Wang, "Configurable GPS/GNSS antenna module resistant to RFI saturation," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 56, no. 1, pp. 381–392, Feb. 2020.
- [61] Q. Mao, L. Huang, Z. Xiang, D. Zhu, and J. Meng, "Nonlinear properties of GaAs/InGaP HBT under high-power microwave pulse injection," *IEEE Trans. Plasma Sci.*, vol. 50, no. 9, pp. 3093–3100, Sep. 2022.
- [62] Analog Devices Inc. (2025). *Product Classification List*. [Online]. Available: <https://www.analog.com/cn/product-category.html>
- [63] Y. T. Lo and S. Lee, *Antenna Handbook: Theory, Applications, and Design*. Cham, Switzerland: Springer, 2013.

- [64] Marki Microwave Inc. (2025). *MM1-1140HCH-2 GaAs Double-Balanced Mixer Datasheet*. [Online]. Available: <https://markimicrowave.com/assets/492dabb9-c6d0-4914-a20b-c006aaf771d4>
- [65] *MCP37231/21-200, MCP37D31/21-200 Datasheet*, Microchip Technology Inc., 2025. [Online]. Available: <https://assets.rs-online.com/v1699464064/Datasheets/69d347eb1c7d6c90712951fc83a2e5f0>
- [66] Texas Instruments Inc. (2025). *ADS1278-SP Radiation Hardened 8-Ch Simultaneous-Sampling 24-Bit Analog-to-Digital Converter Datasheet (Rev. B)*. [Online]. Available: <https://www.ti.com/lit/ds/symlink/ads1278-sp.pdf>
- [67] Y. Sun, P. Babu, and D. P. Palomar, "Majorization-minimization algorithms in signal processing, communications, and machine learning," *IEEE Trans. Signal Process.*, vol. 65, no. 3, pp. 794–816, Feb. 2017.
- [68] G. Heinig and K. Rost, "Representations of toeplitz-plus-hankel matrices using trigonometric transformations with application to fast matrix-vector multiplication," *Linear Algebra Appl.*, vols. 275–276, pp. 225–248, May 1998.
- [69] J.-C. Chen, "Hybrid beamforming with discrete phase shifters for millimeter-wave massive MIMO systems," *IEEE Trans. Veh. Technol.*, vol. 66, no. 8, pp. 7604–7608, Aug. 2017.
- [70] X. Zhang, Z. He, B. Liao, X. Zhang, Z. Cheng, and Y. Lu, "A²RC: An accurate array response control algorithm for pattern synthesis," *IEEE Trans. Signal Process.*, vol. 65, no. 7, pp. 1810–1824, Jul. 2017.



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